Tabel cu derivatele funcţiilor elementare

Funcţia	Derivata	Domeniul de derivabilitate	Funcţia compusă	Derivata
c (constantă)	0	$\mathbb{R}$		
x	1	$\mathbb{R}$	u	u'
$x^n\ (n\in\mathbb{N}^*)$	nx^{n-1}	$\mathbb{R}$	$u^n\ (n\in\mathbb{N}^*)$	$n\cdot u^{n-1}\cdot u'$
$x^\alpha\ (\alpha\in\mathbb{R})$	$\alpha x^{\alpha-1}$	$D_{f'}\supseteq(0,\infty)$	$u^\alpha\ (\alpha\in\mathbb{R},\ u>0)$	$\alpha u^{\alpha-1}\cdot u'$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\mathbb{R}^*$	$\frac{1}{u}\ (u\neq 0)$	$-\frac{u'}{u^2}$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$(0,\infty)$	$\sqrt{u}\ (u>0)$	$\frac{u'}{2\sqrt{u}}$
e^x	e^x	$\mathbb{R}$	e^u	$e^u\cdot u'$
$a^x, 0<a\neq 1$	$a^x\ln a$	$\mathbb{R}$	$a^u\ (0<a\neq 1)$	$a^u\ln a\cdot u'$
$\ln x$	$\frac{1}{x}$	$(0,\infty)$	$\ln u\ (u>0)$	$\frac{u'}{u}$
$\log_a x, 0<a\neq 1$	$\frac{1}{x\ln a}$	$(0,\infty)$	$\log_a u\ (u>0)$	$\frac{u'}{u\ln a}$
$\sin x$	$\cos x$	$\mathbb{R}$	$\sin u$	$\cos u\cdot u'$
$\cos x$	$-\sin x$	$\mathbb{R}$	$\cos u$	$-\sin u\cdot u'$
$\operatorname{tg} x$	$\frac{1}{\cos^2 x}$	$\cos x\neq 0$	$\operatorname{tg} u\ (\cos u\neq 0)$	$\frac{u'}{\cos^2 u}$
$\operatorname{ctg} x$	$-\frac{1}{\sin^2 x}$	$\sin x\neq 0$	$\operatorname{ctg} u\ (\sin u\neq 0)$	$-\frac{u'}{\sin^2 u}$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$(-1,1)$	$\arcsin u\ (\lvert u\rvert\leq 1)$	$\frac{u'}{\sqrt{1-u^2}}$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	$(-1,1)$	$\arccos u\ (\lvert u\rvert\leq 1)$	$-\frac{u'}{\sqrt{1-u^2}}$
$\operatorname{arctg} x$	$\frac{1}{1+x^2}$	$\mathbb{R}$	$\operatorname{arctg} u$	$\frac{u'}{1+u^2}$
$\operatorname{arcctg} x$	$-\frac{1}{1+x^2}$	$\mathbb{R}$	$\operatorname{arcctg} u$	$-\frac{u'}{1+u^2}$
$\operatorname{sh} x=\frac{e^x-e^{-x}}{2}$ (sinus hiperbolic)	$\frac{e^x+e^{-x}}{2}=\operatorname{ch} x$	$\mathbb{R}$	$\operatorname{sh} u$	$\operatorname{ch} u\cdot u'$
$\operatorname{ch} x=\frac{e^x+e^{-x}}{2}$ (cosinus hiperbolic)	$\frac{e^x-e^{-x}}{2}=\operatorname{sh} x$	$\mathbb{R}$	$\operatorname{ch} u$	$\operatorname{sh} u\cdot u'$

Tabel de integrale nedefinite

Nr.crt.	FUNCŢIA	INTEGRALA NEDEFINITĂ
1	$g:\mathbb{R}\to\mathbb{R}, g(x)=x^n, n\in\mathbb{N}$ (funcţia putere cu exponent natural)	$\int x^n \mathrm{d}x = \frac{x^{n+1}}{n+1}+\mathcal{C}$
2	$g: I\to\mathbb{R}$, $I\subset(0,\infty)$, $g(x)=x^a$, $a\in\mathbb{R}-\{-1\}$ (funcţia putere cu exponent real diferit de -1)	$\int x^a \mathrm{d}x = \frac{x^{a+1}}{a+1}+\mathcal{C}$
3	$g: I\to\mathbb{R}$, $I\subset(0,\infty)$ sau $I\subset(-\infty,0)$, $g(x)=\frac{1}{x}=x^{-1}$	$\int \frac{\mathrm{d}x}{x} = \ln\lvert x\rvert+\mathcal{C} = \begin{cases}\ln x, & x>0\\ \ln(-x), & x<0\end{cases}+\mathcal{C}$
4	$g:\mathbb{R}\to\mathbb{R}, g(x)=a^x, a>0$, $a\neq 1$ (funcţia exponenţială de bază a). Dacă $a=e$, atunci $g(x)=e^x$	$\int a^x \mathrm{d}x = \frac{a^x}{\ln a}+\mathcal{C}$ $\int e^x \mathrm{d}x = e^x+\mathcal{C}$
5	$g: I\to\mathbb{R}$, $I=(-\infty,-a)$ sau $I=(-a,a)$, $a>0$ sau $I=(a,\infty)$, $g(x)=\frac{1}{x^2-a^2}$	$\int \frac{\mathrm{d}x}{x^2-a^2} = \frac{1}{2a}\ln\left\lvert\frac{x-a}{x+a}\right\rvert+\mathcal{C}$
6	$g:\mathbb{R}\to\mathbb{R}, g(x)=\frac{1}{x^2+a^2}$, $a\neq 0$	$\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\operatorname{arctg}\frac{x}{a}+\mathcal{C}$
7	$g: I\to\mathbb{R}$, $I\subset(-a,a)$, $a\neq 0$, $g(x)=\frac{1}{\sqrt{a^2-x^2}}$	$\int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a}+\mathcal{C}$
8	$g:\mathbb{R}\to\mathbb{R}$, $g(x)=\frac{1}{\sqrt{x^2+a^2}}$, $a\neq 0$	$\int \frac{\mathrm{d}x}{\sqrt{x^2+a^2}} = \ln(x+\sqrt{x^2+a^2})+\mathcal{C}$
9	$g: I\to\mathbb{R}$, $I\subset(-\infty,-a)$, $I\subset(a,\infty)$, $a>0$, $g(x)=\frac{1}{\sqrt{x^2-a^2}}$	$\int \frac{\mathrm{d}x}{\sqrt{x^2-a^2}} = \ln\lvert x+\sqrt{x^2-a^2}\rvert+\mathcal{C}$

Nr.crt.	FUNCŢIA	INTEGRALA NEDEFINITĂ
10	$g:\mathbb{R}\to\mathbb{R},\ g(x)=\sin x$	$\int \sin x\mathrm{d}x=-\cos x+\mathcal{C}$
11	$g:\mathbb{R}\to\mathbb{R},\ g(x)=\cos x$	$\int \cos x\mathrm{d}x=\sin x+\mathcal{C}$
12	$g:I\to\mathbb{R},$ $I\subseteq\left((2k+1)\frac{\pi}{2},(2k+3)\frac{\pi}{2}\right),$ $k\in\mathbb{Z},\ g(x)=\frac{1}{\cos^2 x}$	$\int \frac{\mathrm{d}x}{\cos^2 x}=\mathrm{tg}x+\mathcal{C}$
13	$g:I\to\mathbb{R},$ $I\subseteq(k\pi,(k+1)\pi),\ k\in\mathbb{Z},$ $g(x)=\frac{1}{\sin^2 x}$	$\int \frac{\mathrm{d}x}{\sin^2 x}=-\mathrm{ctg}x+\mathcal{C}$
14	$g:I\to\mathbb{R},$ $I\subseteq\left((2k+1)\frac{\pi}{2},(2k+3)\frac{\pi}{2}\right),$ $k\in\mathbb{Z},\ g(x)=\mathrm{tg}x$	$\int \mathrm{tg}x\mathrm{d}x=-\ln\lvert\cos x\rvert+\mathcal{C}$
15	$g:I\to\mathbb{R},$ $I\subseteq(k\pi,(k+1)\pi),\ k\in\mathbb{Z}$ $g(x)=\mathrm{ctg}x$	$\int \mathrm{ctg}x\mathrm{d}x=\ln\lvert\sin x\rvert+\mathcal{C}$
16	$g:\mathbb{R}\to\mathbb{R},$ $g(x)=\frac{1}{2}(e^x-e^{-x})=\mathrm{sh}\ x$ (funcţia sinus hiperbolic)	$\int \mathrm{sh}\ x\mathrm{d}x=\mathrm{ch}\ x+\mathcal{C}$
17	$g:\mathbb{R}\to\mathbb{R},$ $g(x)=\frac{1}{2}(e^x+e^{-x})=\mathrm{ch}\ x,$ (funcţia cosinus hiperbolic)	$\int \mathrm{ch}\ x\mathrm{d}x=\mathrm{sh}\ x+\mathcal{C}$

Observaţie. Probaţi că funcţia G, care apare în integrala nedefinită, este primitivă pentru g, verificând egalitatea $G'=g$.